Homework - 1

Page No. :
Date :

Q4

$$H(X,Y) = -\sum_{x,y} p(x,y)\log(p(x,y))$$

$$H(X) = -\sum_{x} p(x)\log(p(x))$$

$$H(Y|X) = -\sum_{x,y} p(x,y)\log p(y|x)$$

Proof

$$p(x,y) = p(x) \cdot p(y|x)$$

taking log

$$\log(p(x,y)) = \log p(x) + \log(p(y|x))$$

$$H(X,Y) = -\sum_{x,y} p(x,y)\log(p(x,y))$$

$$= -\sum_{x,y} p(x,y)\left(\log p(x) + \log p(y|x)\right)$$

$$= -\sum_{x,y} p(x,y)\log(p(x)) - \sum_{x,y} p(x,y)\log(p(y|x))$$

$$= -\sum_{x}\left[\sum_{y} p(x,y)\log(p(x))\right] - \sum_{x,y} p(x,y)\log(p(y|x))$$

$$= -\sum_{x} p(x)\log(p(x)) + H(Y|X)$$

Page No. :
Date :

$$= H(X) + H(Y|X)$$

Hence proved

Q5

$$I(X;Y) = \sum p(x,y) \log \left[ \frac{p(x,y)}{p(x) \cdot p(y)} \right]$$

(a) $I(X;Y) = H(X) - H(X|X)$

$$I(X;Y) = \sum p(x,y) \log \left[ \frac{p(x,y)}{p(x)\, p(y)} \right]$$

$$= \sum p(x,y) \log (p(x,y))$$

$$- \sum p(x,y) \log p(x)$$

$$- \sum p(x,y) \log p(y)$$

$$= \sum p(x,y) \log (p(x,y)) - \sum_x \sum_y p(x,y) \log p(x)$$

$$- \sum_y \sum_x p(x,y) \log p(y)$$

$$= \sum p(x,y) \log (p(x,y)) - \sum_x p(x) \log p(x)$$

$$- \sum_y p(y) \log p(y)$$

$$= \sum p(x,y) \log (p(x,y)) + H(X) + H(Y)$$

* $$p(x,y) = p(y)\, p(x|y)$$

$$\log p(x,y) = \log(p(y)) + \log(p(x|y))$$

$$= \sum p(x,y)\left(\log p(y) + \log p(x|y)\right) + H(X) + H(Y)$$

$$= \sum p(x,y) \log (p(y)) + \sum p(x,y) \log p(x|y) + H(X) + H(Y)$$

$$= \sum_{y} \sum_{x} p(x,y) \log (p(y)) + \sum p(x,y) \log p(x|y) + H(X) + H(Y)$$

$$= \sum_{y} p(y) \log (p(y)) + \sum p(x,y) \log p(x|y) + H(X) + H(Y)$$

$$= -H(Y) - H(X|Y) + H(X) + H(Y)$$

$$= H(X) - H(X|Y)$$

(b) $I(X;Y) = H(Y) - H(Y|X)$

$$I(X;Y) = \sum p(x,y) \log\left(\frac{p(x,y)}{p(x),p(y)}\right)$$

$$= \sum p(x,y) \log p(x,y) - \sum p(x,y) \log p(x)$$

$$- \sum p(x,y) \log p(y)$$

$$= \sum p(x,y) \log p(x,y) - \sum_x \sum_y p(x,y) \log p(x)$$

$$- \sum_y \sum_x p(x,y) \log p(y)$$

$$= \sum p(x,y) \log p(x,y) - \sum p(x) \log p(x)$$

$$- \sum p(y) \log p(y)$$

$$= \sum p(x,y) \log p(x,y) + H(X) + H(Y)$$

* $p(x,y) = p(x) . p(y|x)$

$$\log(p(x,y)) = \log p(x) + \log p(y|x)$$

$$= \sum p(x,y) \log p(x) + \sum p(x,y) \log (y|x)$$

$$+ H(X) + H(Y)$$

$$= \sum_x \sum_y p(x,y) \log p(x) - H(Y|X) + H(X) + H(Y)$$

$$= \sum_x p(x) \log p(x) - H(Y|X) + H(X) + H(Y)$$

$$= -H(X) - H(Y|X) + H(X) + H(X)$$

$$= H(Y) - H(Y|X)$$

Q6

$$p(x=0, y=0) = p(x=0, y=1) = p(x=1, y=1) = 1/3$$

$$p(x=1, y=0) = 0$$

| $x$ \ $y$ | 0 | 1 |
|---|---|---|
| 1 | 0 | 1/3 |
| 0 | 1/3 | 1/3 |

(a) $p(x=0) = p(x=0, y=0) + p(x=0, y=1)$

$$= 1/3 + 1/3$$

$$= 2/3$$

$p(x=1) = p(x=1, y=0) + p(x=1, y=1)$

$$= 0 + 1/3$$

$$= 1/3$$

$$H(x) = -\Sigma p(x) \log p(x)$$

$$= -\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log\left(\frac{1}{3}\right)$$

$$= 0.918$$

$$p(y=0) = p(x=0, y=0) + p(x=1, y=0)$$
$$= 1/3 + 0$$
$$= 1/3$$

$$p(y=1) = p(x=0, y=1) + p(x=1, y=1)$$
$$= 1/3 + 1/3$$

$$= 2/3$$

$$H(Y) = -\Sigma p(y) \log p(y)$$

$$= -\frac{1}{3}\log_2(1/3) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)$$

$$= 0.918$$

(b)

$$H(X,Y) = -\Sigma p(x,y) \log p(x,y)$$

$$= -\left(0 \times \log(0) + \frac{1}{3}\log\left(\frac{1}{3}\right) + \frac{1}{3}\log\left(\frac{1}{3}\right) + \frac{1}{3}\log\left(\frac{1}{3}\right)\right)$$

$$= -3 \times \frac{1}{3}\log_2\left(\frac{1}{3}\right)$$

$$= -\log_2\left(\frac{1}{3}\right)$$

$$= \log_2(3)$$

$$= 1.585$$

$$H(Y/X) = H(x,y) - H(x)$$
$$= 1.585 - 0.918$$
$$= 0.667$$

$$H(X|Y) = H(x,y) - H(Y)$$
$$= 1.585 - 0.918$$
$$= 0.667$$

(c)

$$H(x,y) = -\sum p(x,y) \log p(x,y)$$

$$= -\left(0 \times \log(0) + \frac{1}{3}\log\left(\frac{1}{3}\right) + \frac{1}{3}\log\left(\frac{1}{3}\right) + \frac{1}{3}\log\left(\frac{1}{3}\right)\right)$$

$$= -3 \times \frac{1}{3}\log_2\left(\frac{1}{3}\right)$$

$$= 1.585$$

Page No.:
Date:

(d)

$H(Y) - H(Y|X)$

$\Rightarrow 0.918 - 0.667$

$\Rightarrow 0.251$

(e)

$I(X;Y) = H(Y) - H(Y|X)$

$= 0.918 - 0.667$

$= 0.251$